

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q24-2452

Kinsington 100 kV  
September 02, 2025

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2452. The Interconnection Customer proposes a 200.0 MW solar photovoltaic facility interconnecting at the Kinsington 100 kV substation in the Illinois area. The request was received on September 17, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 469 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,024,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

Kinsington-River Rock 100 kV: 124.0% loading (140 MVA) following the outage of the Kinsington-Jacobs Corner 100 kV; DFAX = 100.0%.

Kinsington Substation-Jacobs Corner 100 kV: 122.3% loading (142 MVA) following the outage of the Kinsington-River Rock 100 kV; DFAX = 100.0%.

The Cypress Creek-Big Oaks 100 kV is overloaded to 105.5% of its 500 MVA emergency rating for the loss of the Dekalb-Jones Chapel 500 kV. The project contributes 5.2% DFAX (10.4 MW) to this overload.

Big Oaks-Laureles 100 kV: 104.5% loading (531 MVA) following the outage of the Dekalb-Jones Chapel 500 kV; DFAX = 5.2%.

The Ward Gulf Substation-Laureles 100 kV is overloaded to 103.7% of its 562 MVA emergency rating for the loss of the Jones Chapel-Dekalb Substation 500 kV. The project contributes 5.2% DFAX (10.4 MW) to this overload.

For the loss of Dekalb-Richland 500 kV, the Waynedale-Montpelier Electric Generating Station 230 kV loads to 103.6% of its 1,919 MVA rating. Project DFAX: 20.4%.

The Minnehaha-Sidney A Murray Jr Hydroelectric 230 kV is overloaded to 103.0% of its 2,041 MVA emergency rating for the loss of the Richland-Dekalb Substation 500 kV. The project contributes 16.0% DFAX (32.0 MW) to this overload.

Pickford-Herald-Twin Cities 230 kV: 102.5% loading (2,114 MVA) following the outage of the Trego-Beaver 500 kV; DFAX = 9.4%.

For the loss of Dekalb-Jones Chapel 500 kV Line, the Ege-Golden Gate 100 kV loads to 102.4% of its 398 MVA rating. Project DFAX: 5.2%.

The Post Rock-Myrtle Pt 230 kV is overloaded to 102.0% of its 2,995 MVA emergency rating for the loss of the Beaver-Trego 500 kV line. The project contributes 12.4% DFAX (24.7 MW) to this overload.

Newport Tie-Post Rock Substation 230 kV line: 102.0% loading (2,963 MVA) following the outage of the Trego-Beaver 500 kV; DFAX = 12.8%.

Weedpatch-Canal Rd 230 kV: 102.0% loading (1,782 MVA) following the outage of the Trego-Beaver 500 kV; DFAX = 7.4%.

The Cuba City-Ege 100 kV L is overloaded to 101.8% of its 414 MVA emergency rating for the loss of the Dekalb-Jones Chapel 500 kV. The project contributes 5.2% DFAX (10.4 MW) to this overload.

For the loss of Lerdo-Tidd 500 kV L, the Tidd-Filer 500 kV loads to 101.5% of its 2,413 MVA rating. Project DFAX: 6.9%.

For the loss of Ritter Park-Sugar Hill 345 kV, the Medora-P.T.S 345 kV loads to 101.4% of its 2,278 MVA rating. Project DFAX: 7.4%.

The Golden Gate-Laotto 100 kV Line is overloaded to 101.0% of its 385 MVA emergency rating for the loss of the Jones Chapel Substation-Dekalb 500 kV. The project contributes 5.2% DFAX (10.4 MW) to this overload.

For the loss of Dekalb-Jones Chapel 500 kV L, the Farmersville-Cuba City 100 kV loads to 100.6% of its 448 MVA rating. Project DFAX: 5.2%.

Kendallville Substation-Rex Brown 100 kV Line: 100.3% loading (487 MVA) following the outage of the Dekalb-Jones Chapel 500 kV; DFAX = 5.2%.

The Rex Brown-Montibello 100 kV is overloaded to 100.2% of its 506 MVA emergency rating for the loss of the Jones Chapel-Dekalb 500 kV L. The project contributes 5.2% DFAX (10.4 MW) to this overload.

Ligonier-Montiello 100 kV line: 100.2% loading (519 MVA) following the outage of the Dekalb-Jones Chapel 500 kV; DFAX = 5.2%.

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                                | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Kinsington-River Rock 100 kV                            | Rebuild line with higher-capacity conductor | \$3,669,000   | 100% (\$3,669,000)    |
| Kinsington Substation-Jacobs Corner 100 kV              | Rebuild line with higher-capacity conductor | \$6,371,000   | 100% (\$6,371,000)    |
| Big Oaks-Cypress Crk Substation 100 kV                  | Reconductor line                            | \$3,636,000   | 2% (\$73,000)         |
| Big Oaks-Laureles 100 kV                                | Reconductor line                            | \$1,007,000   | 2% (\$20,000)         |
| Ward Gulf-Laureles 100 kV                               | Reconductor line                            | \$2,891,000   | 2% (\$58,000)         |
| Waynedale-Montpelier Electric Generating Station 230 kV | Reconductor line                            | \$16,070,000  | 2% (\$324,000)        |
| Minnehaha-Sidney A Murray Jr Hydroelectric 230 kV       | Reconductor line                            | \$10,828,000  | 2% (\$218,000)        |
| Pickford-Herald-Twin Cities 230 kV                      | Reconductor line                            | \$17,807,000  | 2% (\$359,000)        |
| Ege-Golden Gate 100 kV L                                | Reconductor line                            | \$1,009,000   | 2% (\$20,000)         |
| Post Rock-Myrtle Pt 230 kV                              | Reconductor line                            | \$10,415,000  | 2% (\$210,000)        |
| Newport Tie-Post Rock Substation 230 kV line            | Reconductor line                            | \$12,155,000  | 2% (\$245,000)        |
| Weedpatch-Canal Rd 230 kV                               | Reconductor line                            | \$4,629,000   | 2% (\$93,000)         |

| Facility                       | Required Upgrade           | Cost Estimate | Allocation to Project |
|--------------------------------|----------------------------|---------------|-----------------------|
| Ege-Cuba City 100 kV           | Reconductor line           | \$2,138,000   | 2% (\$43,000)         |
| Tidd-Filer 500 kV              | Reconductor line           | \$37,685,000  | 2% (\$767,000)        |
| Medora-P.T.S 345 kV            | Reconductor line           | \$14,969,000  | 2% (\$303,000)        |
| Golden Gate-Laotto 100 kV Line | Reconductor line           | \$1,875,000   | 2% (\$38,000)         |
| Farmersville-Cuba City 100 kV  | Reconductor line           | \$1,832,000   | 2% (\$37,000)         |
| Kendallville-Rex Brown 100 kV  | Reconductor line           | \$1,424,000   | 2% (\$29,000)         |
| Rex Brown-Montibello 100 kV    | Reconductor line           | \$5,652,000   | 2% (\$114,000)        |
| Ligonier-Montiello 100 kV line | Reconductor line           | \$1,641,000   | 2% (\$33,000)         |
|                                | Total allocated to project |               | \$13,024,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.